

TEAM 3

Hospital Information System – CarePlus Hospital

Done by

Kishan Ram (192421268)

Vimal A (192421401)

Ajay U (192421237)

Adhithyan S (192421102)

1. Security Risks

Risk	Description
Unauthorized Access	Attackers gain access to patient records.
Data Breach	Sensitive patient information may be stolen.
Weak Password Security	Poor password storage leads to credential theft.
Insider Threat	Staff using unmanaged devices expose data.
Malware/Ransomware	Unsecured devices introduce malware.
Regulatory Non-Compliance	Violates healthcare privacy regulations.
Service Downtime	Cyberattacks disrupt the portal.

2. Vulnerabilities

Vulnerability	Description	Severity
Weak Password Storage	Passwords stored insecurely	High
No Security Reviews	Security testing skipped	High
Unmanaged Devices	Personal devices access system	High
Missing MFA	Password-only authentication	High
Delayed Patching	Known flaws remain	Medium
Poor Access Control	Excessive permissions	Medium
Lack of Security Training	Users vulnerable to phishing	Medium
Weak Logging	Attacks may go unnoticed	Low

3. Cyber Threats

Threat	Impact
Phishing	Credential theft
Ransomware	Encrypts hospital data
SQL Injection	Steals/modifies records
Cross-Site Scripting	Session hijacking
Insider Attack	Misuse of privileges
Brute Force	Account compromise
DoS	Portal unavailable
Malware	System corruption

4. DevOps vs DevSecOps

DevOps	DevSecOps
Fast software delivery	Secure software delivery
Security at end	Security throughout SDLC
Separate security team	Shared responsibility
Manual testing	Automated security testing
Limited scanning	Continuous scanning
Late issue detection	Early issue detection
Compliance after deployment	Compliance built into CI/CD
Focus on speed	Focus on speed + security

5. OWASP Best Practices

- Use strong password hashing (bcrypt, Argon2, PBKDF2).
- Enable Multi-Factor Authentication (MFA).
- Validate all user inputs to prevent SQL Injection and XSS.
- Apply the Principle of Least Privilege.
- Perform regular vulnerability scanning and penetration testing.

6. Business Impact

- Data breaches and privacy loss
- Financial losses and legal penalties
- Loss of patient trust
- Operational disruption
- Service downtime
- Regulatory non-compliance

7. Concept Map

Risk → Unauthorized Access



Vulnerability → Weak Password Storage



DevSecOps Practice → Strong Hashing + MFA + Security Testing



Business Benefit → Privacy, Compliance, Trust, Availability